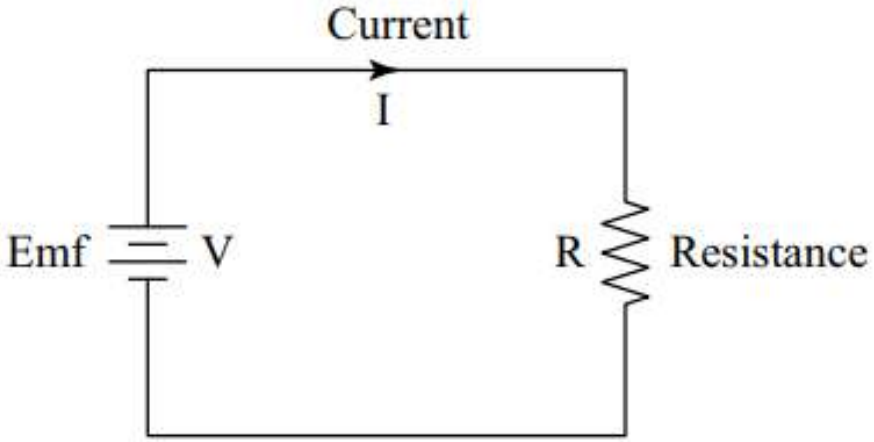
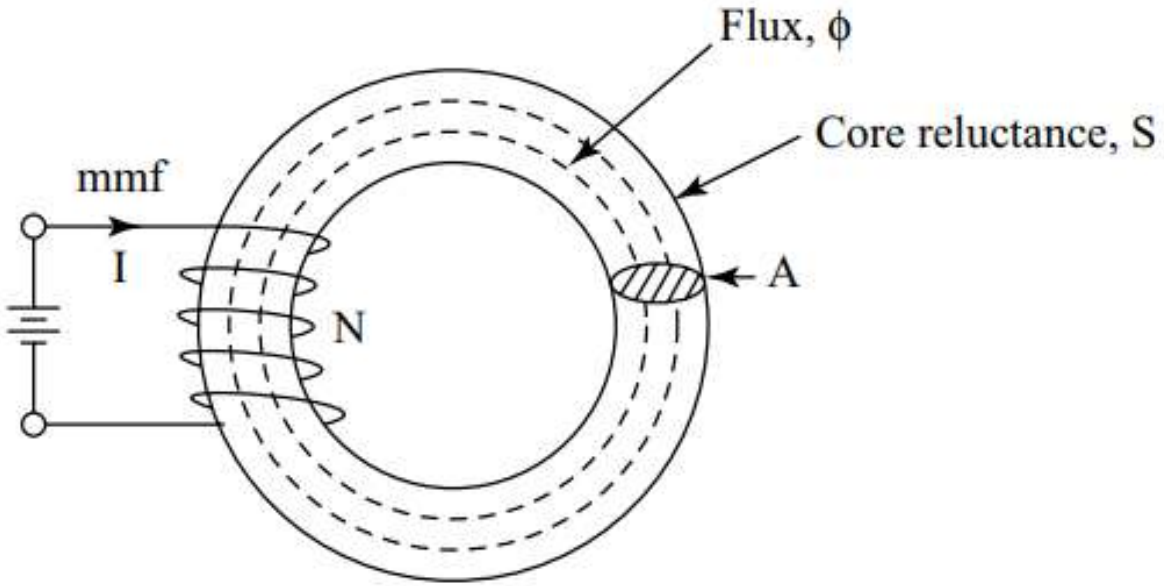


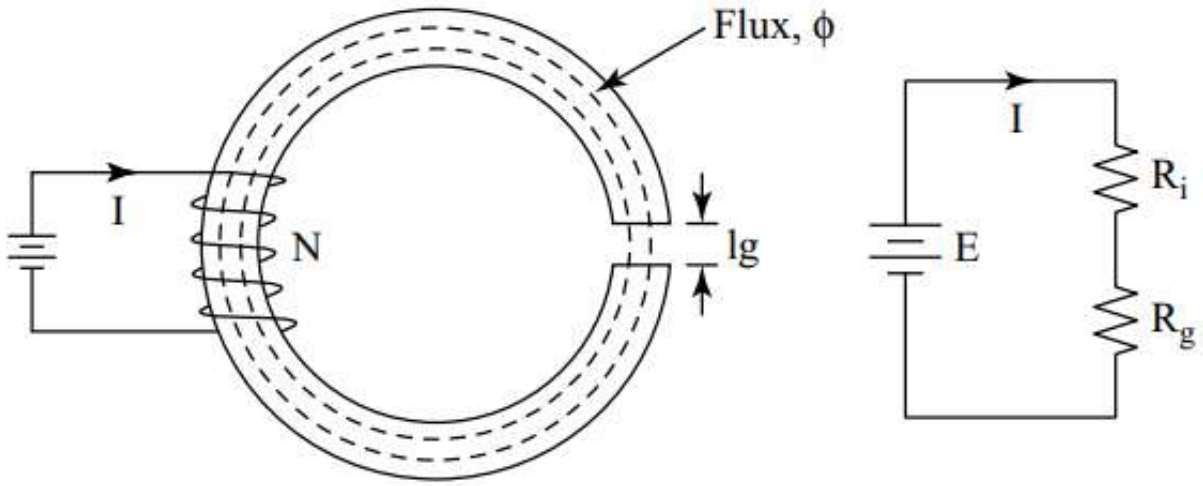
Electric Circuit



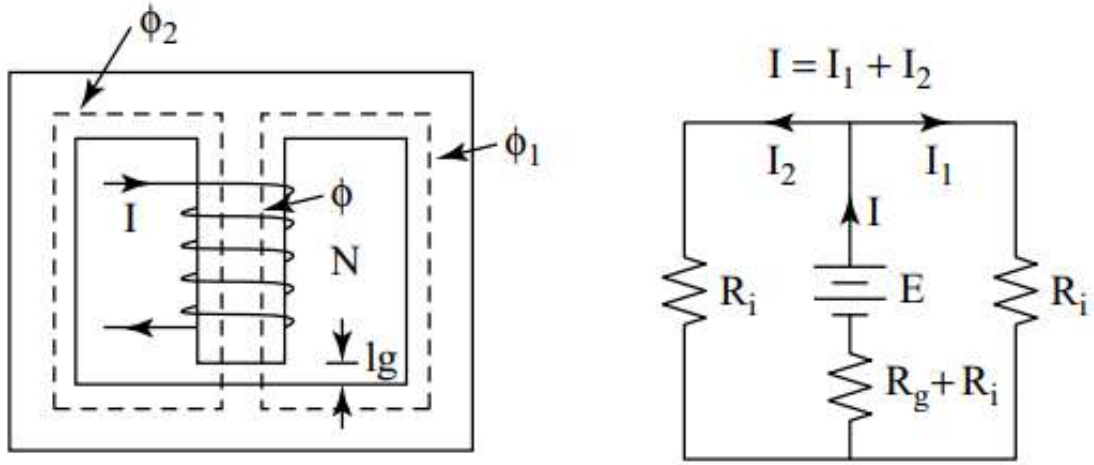
Magnetic Circuit

Table 5.1 *Comparison Between an Electric Circuit and a Magnetic Circuit*

Electric Circuit	Magnetic Circuit
Current, I	Flux, ϕ
EMF, V	MMF, N I
Resistance, R	Reluctance, S
$I = \frac{EMF}{Resistance} = \frac{V}{R}$	$\phi = \frac{MMF}{Reluctance} = \frac{NI}{S}$
$R = \rho \frac{l}{a}$	$S = \frac{l}{\propto A}$
$Conductance = \frac{1}{Resistance}$	$Permeance = \frac{1}{Reluctance}$
$Electric\ field\ intensity = \frac{V}{l}$	$Magnetic\ field\ intensity,\ H = \frac{AT}{l}$ or, $H = \frac{NI}{l}$



(a) Series circuit



(b) Parallel circuit

Figure 5.18 *Series and parallel magnetic circuits with equivalent electric circuits*

Example 5.2 A circular iron ring of mean diameter 25 cm and cross-sectional area 9 cm^2 is wound with a coil of 100 turns and carries a current of 1.5 A. The relative permeability of iron is 2000. Calculate the amount of flux produced in the ring.

Solution:

Mean length of flux path, $\ell = \pi D = 3.14 \times 25 \text{ cm} = \dfrac{3.14 \times 25}{100} \text{ m} = 0.785 \text{ m}$

Flux, $$\phi = \frac{\text{MMF}}{\text{Reluctance}} = \frac{\text{NI}}{\ell / \mu_o \mu_r A} = \frac{100 \times 1.5}{0.785 / 4\pi \times 10^{-7} \times 2000 \times 9 \times 10^{-4}}$$

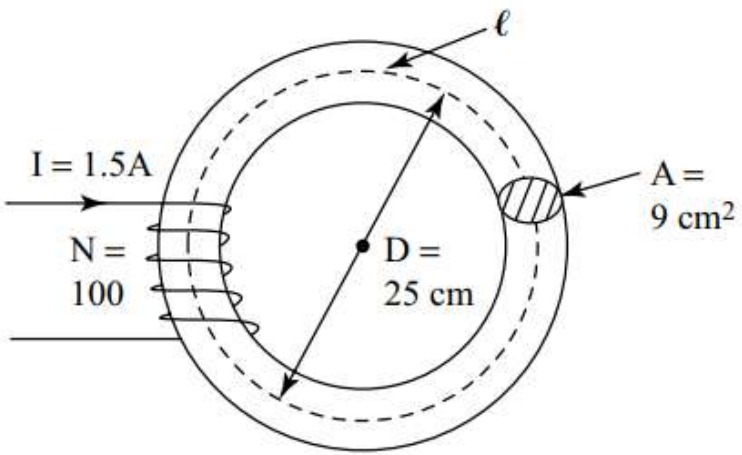


Figure 5.21

$$= \frac{150}{348 \times 10^3} = \frac{150}{348} \times 10^{-3}$$

$$= 0.431 \times 10^{-3} \text{ Wb}$$

$$= 0.431 \text{ mWb}$$

Example 5.3 A rectangular shape iron core has an air gap of 0.01 cm. The mean length of the flux path through iron is 39.99 cm. The relative permeability of iron is 2000. The coil has 1000 turns. The cross-sectional area of the core is 9 cm^2. Calculate the current required to produce a flux of 1 mWb in the core.

Solution:

Total reluctance of the flux path = Reluctance of iron path + Reluctance of air gap,

i.e.,
$$S = S_i + S_g$$
$$S = \frac{l_i}{\propto_o \propto_r A} + \frac{l_g}{\propto_o A}$$

Note: the iron path permeability is μ which is equal to $\mu_0\mu_r$ whereas for the air gap the permeability is μ_0 only.

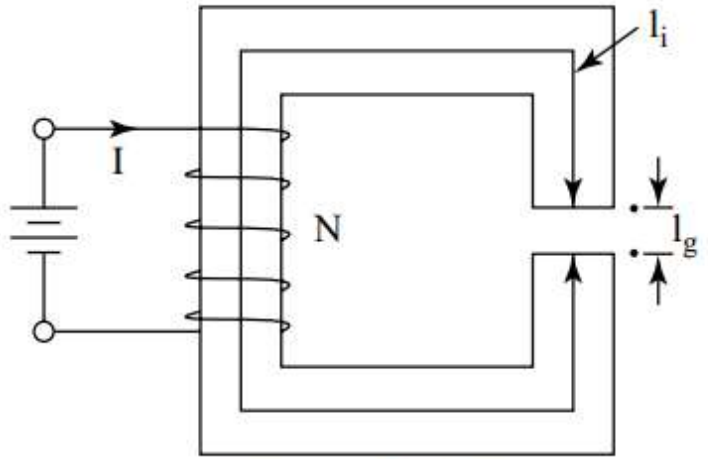


Figure 5.22

Substituting the given values,

$$S = \frac{39.09 \times 10^{-2}}{4\pi \times 10^{-7} \times 2000 \times 9 \times 10^{-4}} + \frac{0.01 \times 10^{-2}}{4\pi \times 10^{-7} \times 9 \times 10^{-4}}$$

$$= \frac{10^6}{4\pi}\left[\frac{39.09}{18} + \frac{100}{10}\right] = \frac{295.45 \times 10^5}{36\pi}$$

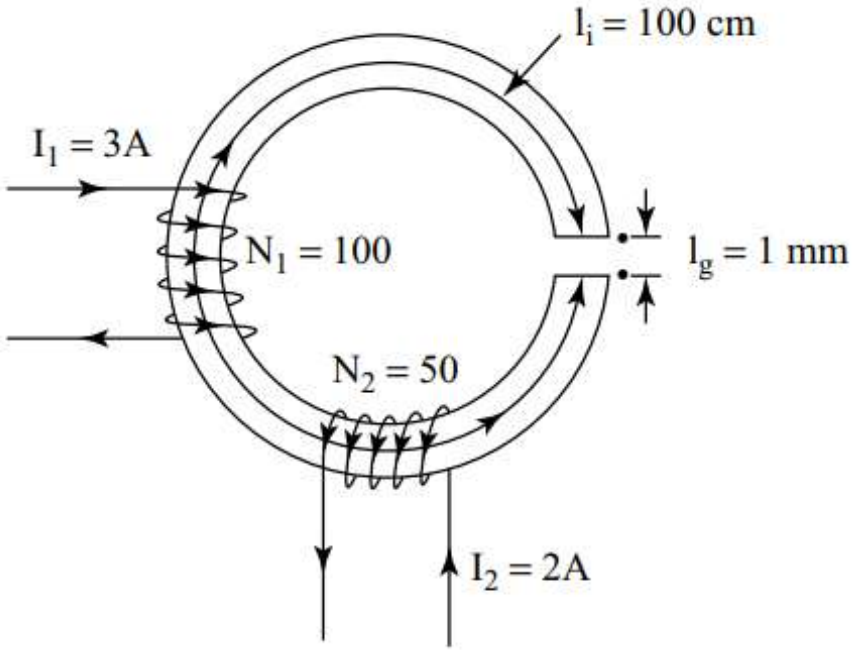
Flux,

$$\phi = \frac{NI}{S} = \frac{1000\ I}{S}$$

$\therefore$

$$I = \frac{\phi \times S}{1000} = \frac{1 \times 10^{-3} \times 295.45 \times 10^5}{36\pi \times 1000} = \frac{29.545}{36\pi} = 0.26\text{A}$$

Example 5.5 An iron ring of mean length of an iron path of 100 cm and having a uniform cross-sectional area of 10 cm^2 is wound with two magnetizing coils as shown. The direction of current flowing through the two coils are such that they produce flux in the opposite directions. The permeability of iron is 2000. There is a cut in the ring creating an air gap of 1 mm. Calculate the flux available in the air gap.



The net MMF will be the resultant effect of MMF of the two coils in producing the flux in the core. As in Fig. 5.25, the flux produced by the MMF of the two coils are in opposite directions. Thus, the resultant MMF will be the difference of these two MMFs.

Resultant MMF $= N_1 I_1 - N_2 I_2$

$= 100 \times 3 - 50 \times 2$

$= 200$

Total reluctance, $S = S_i + S_g$

$$= \frac{l_i}{\propto_o \propto_r A_i} + \frac{l_g}{\propto_o A_g}$$ [we have considered the reluctance of $\mu_o\mu_r$ for iron and μ_o for air]

Since $A_i = A_g$, i.e., the cross-sectional area of the iron path is the same as that of the air gap,

$$S = \frac{1}{\mu_o A}\left[\frac{l_i}{\mu_r} + l_g\right]$$

Substituting values $$S = \frac{1}{4\pi \times 10^{-7} \times 10 \times 10^{-4}}\left[\frac{50 \times 10^{-2}}{2000} + 1 \times 10^{-3}\right]$$

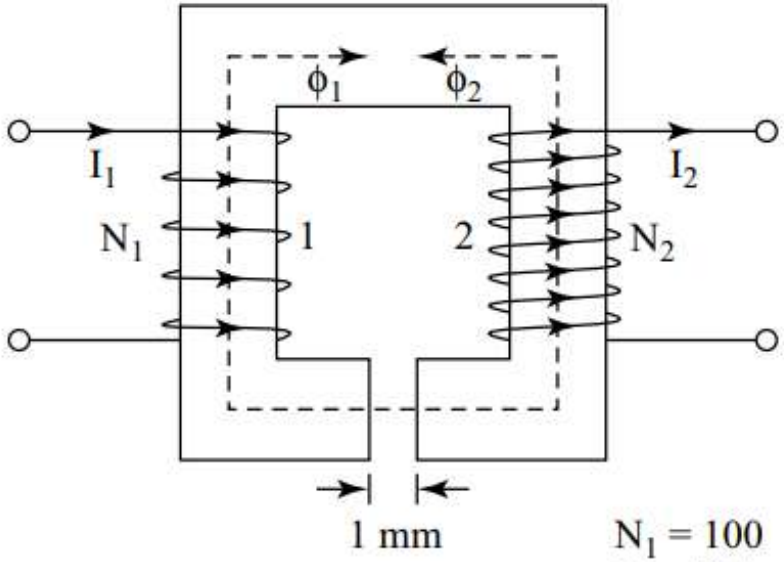
$$S = \frac{10^{10}}{4\pi}[25 \times 10^{-5} + 1 \times 10^{-3}]$$

$$= \frac{10^{10}}{4\pi} 10^{-3} [25 \times 10^{-2} + 1]$$

$$= \frac{10^7}{4\pi}[1.25] = 10^6$$

Flux, $$\phi = \frac{MMF}{S} = \frac{200}{10^6} = 2 \times 10^{-4} = 0.2 \text{ mWb}$$

Example 5.6 Calculate the flux produced in the air gap in the magnetic circuit shown in Fig. 5.26, which is excited by the MMF of two windings. The mean length of the flux path is 40 cm. The permeability of iron is 2000. The uniform core cross-sectional area is 10 cm^2.



$N_1 = 100$
$I_1 = 10$ A
$N_2 = 80$
$I_2 = 105$A

Solution:

Applying Fleming's thumb rule we can see that the ampere turns of coil 2 produce flux in the opposite direction as the flux produced by the ampere turns of coil 1.

Total MMF $= N_1 I_1 - N_2 I_2$

$= 100 \times 10 - 80 \times 1.5$

$= 880$ AT

Total reluctance = Reluctance of iron + Reluctance of air gap

$$= \frac{l_i}{\mu_o \mu_r A} + \frac{l_g}{\mu_o A} = \frac{1}{\mu_o A}\left[\frac{l_i}{\mu_r} + l_g\right]$$

Substituting values

Total Reluctance $$= \frac{1}{4\pi \times 10^{-7} \times 10 \times 10^{-4}}\left[\frac{(40 \times 10^{-2} - 1 \times 10^{-3})}{2000} + 1 \times 10^{-3}\right]$$

$$= \frac{10^{10}}{4\pi}\left[\frac{399 \times 10^{-3}}{2000} + 1 \times 10^{-3}\right]$$

$$= \frac{10^7}{4\pi}\left[\frac{399}{2000} + 1\right]$$

$$= 0.955 \times 10^6 \text{ AT/Wb.}$$

Core Flux $$= \frac{\text{MMF}}{\text{Reluctance}} = \frac{880}{0.955 \times 10^6} = 921.5 \times 10^{-6}$$

$$= 0.9215 \times 10^{-3} \text{ Wb}$$

The air gap flux is the same as the core flux as the whole of the core flux crosses the air gap and there is no fringing.